SENTINELNET – AI-POWERED NETWORK INTRUSION DETECTION SYSTEM (NIDS)

Authors: Upasana Prabhakar
Research Supervisor: Dr. N Jagan Mohan

ABSTRACT: SentinelNet aims to build a smart Network Intrusion Detection System (NIDS) powered by AI and machine learning. The central goal is to automatically identify and respond to suspicious or potentially harmful network traffic and cyber-attacks in real time. Using historical data, machine learning models will be trained to distinguish between normal and malicious activities, extract crucial features, classify threats, and trigger alerts to help network defenders respond quickly and effectively.

KEYWORDS: Intrusion Detection, Cybersecurity, Machine Learning, AI, Network Traffic

STATEMENT OF ORIGINALITY: This paper presents SentinelNet, a hybrid IDS combining signature-based and anomaly-based methods with advanced preprocessing, feature selection, and class-imbalance handling, applied to NSL-KDD and CICIDS2017 datasets.

1 INTRODUCTION

SentinelNet aims to build a smart Network Intrusion Detection System (NIDS) powered by AI and machine learning. The central goal is to automatically identify and respond to suspicious or potentially harmful network traffic and cyber-attacks in real time.

Using historical data, machine learning models will be trained to:

- Distinguish between normal and malicious activities
- Extract crucial features
- Classify threats
- Trigger alerts to help network defenders respond quickly and effectively

2 LITERATURE

The field of Intrusion Detection Systems (IDS) has evolved from rule-based methods to AI-powered frameworks. Literature highlights:

- **Signature-Based IDS:** Relies on predefined attack signatures, effective for known threats but fails on zero-day attacks.
- **Anomaly-Based IDS:** Uses statistical models and ML to detect unusual traffic patterns, can detect unknown threats but may have higher false positives.

2.1 Benchmark datasets shaping IDS research

- KDD Cup 1999: Early dataset, criticized for redundancy and outdated attacks.
- NSL-KDD: Improved version, duplicates removed, balanced dataset for academic evaluation.
- CICIDS2017: Realistic enterprise traffic, modern attacks like DDoS, infiltration, and web exploits.

2.2 ML algorithms widely used

Decision Trees, Random Forests, SVMs, Neural Networks, Deep Learning, Ensemble methods, Hybrid IDS models.

2.3 Insights

- Feature selection and preprocessing are critical.
- Class imbalance is a challenge.
- Hybrid approaches combine speed of signature-based and intelligence of anomaly-based IDS.

3 METHODOLOGY

3.1 Database

3.1.1 NSL-KDD Dataset

A refined dataset to reduce redundancy:

- 41 features: basic (protocol, service, bytes) + traffic-level (flags, errors)
- Training: 125,973 records
- Test: 22,544 records
- Attack types: DoS (smurf, neptune), Probe (portsweep, nmap), U2R (buffer_overflow, rootkit), R2L (ftp_write, guess_passwd)

3.1.2 CICIDS2017 Dataset

Realistic enterprise network traffic:

- 78–83 features: flow, statistical, packet-level
- Over 2.8 million labeled flow records across 5 days
- Attack types: DDoS (Hulk, GoldenEye, Slowloris), Brute Force, Web Attacks (SQLi, XSS), Botnet/Infiltration, Reconnaissance (PortScan, Heartbleed)

3.1.3 Dataset Comparison

Table 1: Dataset Comparison

Attribute	NSL-KDD	CICIDS2017
No. of Features	41	78–83
Total Records	$\sim$150,000	$>$2.8 million
Traffic	Simulated, old-generation	Realistic, enterprise-scale
Usage	Lightweight model evaluation	Practical deployment
Attack Variety	4 legacy categories	14+ modern attacks

3.2 Preprocessing

- Handling missing values
- Feature scaling: MinMaxScaler, StandardScaler
- Encoding categorical features: One-Hot Encoding
- Stratified train-test split to maintain class balance

3.2.1 Scaling vs Normalization

Table 2: Scaling vs Normalization

Aspect	Standardization	Normalization
Definition	Transforms features to mean=0, std=1	Rescales to [0,1]
Purpose	For Gaussian-based algorithms	For bounded-input algorithms
Use Cases	SVM, Logistic Regression, PCA	Neural Nets, KNN
Effect on Outliers	Less sensitive	Sensitive (compressed to 0–1)
Preserves Data Shape	Yes	Yes, but centers differently
When to Use	Gaussian-based models	NN, image data

3.3 Proposed Method: Logistic Regression

Logistic Regression is a model that predicts the probability of a data point belonging to a class using a simple formula:

$$P = \frac{1}{1 + e^{-z}}, \quad \text{where } z = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

Here, x_i are input features, w_i are weights, and b is the bias. The output probability is then converted to a class label (commonly using 0.5 as the threshold). Logistic Regression is simple, interpretable, and works well for linearly separable data, but may underperform for complex non-linear patterns in IDS.

4 RESULTS

4.1 NSL-KDD Results

4.1.1 Model Accuracy Scores

Table 3: NSL-KDD: Accuracy per model

Model	Accuracy (%)
Logistic Regression	77.56
Decision Tree	79.03
Random Forest	80.60
Gradient Boosting	77.95

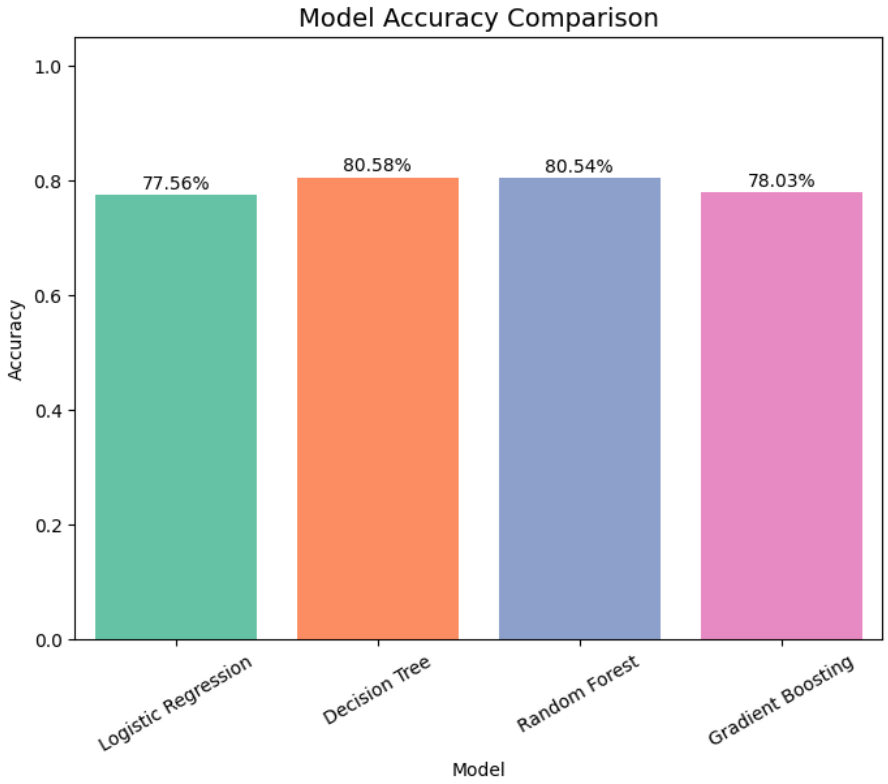

Figure 1: NSL-KDD: Model accuracy comparison.

4.1.2 Macro Average Performance

Table 4: NSL-KDD: Macro Average Precision, Recall, F1

Model	Precision	Recall	F1-score
Logistic Regression	0.80	0.79	0.78
Decision Tree	0.81	0.81	0.79
Random Forest	0.83	0.83	0.81
Gradient Boosting	0.80	0.80	0.78

4.1.3 Weighted Average Performance

Table 5: NSL-KDD: Weighted Average Precision, Recall, F1

Model	Precision	Recall	F1-score
Logistic Regression	0.81	0.78	0.77
Decision Tree	0.83	0.79	0.79
Random Forest	0.85	0.81	0.80
Gradient Boosting	0.82	0.78	0.78

4.1.4 Confusion Matrix Explanation

The confusion matrix provides detailed insights into classification:

Table 6: Structure of a Confusion Matrix

	Predicted Normal	Predicted Attack
Actual Normal	True Negative (TN)	False Positive (FP)
Actual Attack	False Negative (FN)	True Positive (TP)

Interpretation:

- TN: Correctly identified benign traffic.
- FP: Benign traffic misclassified as attack (false alarm).
- FN: Attack traffic missed (dangerous miss).
- TP: Correctly detected attack.

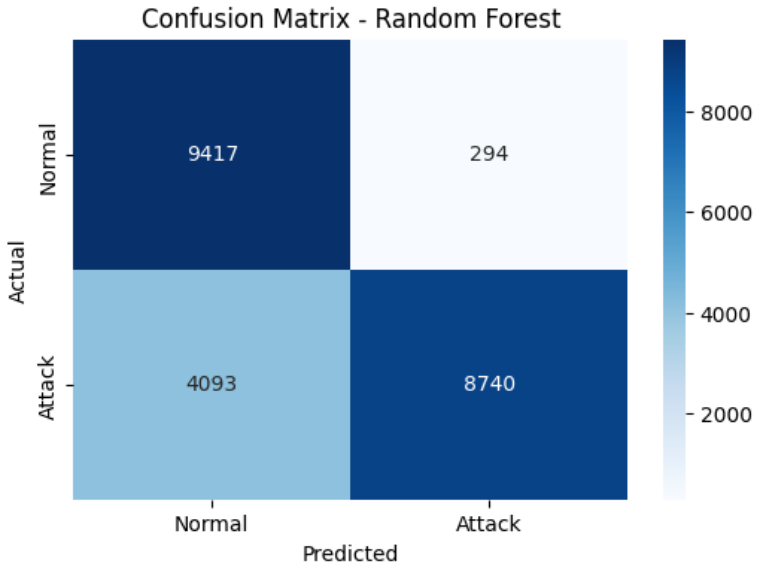

Figure 2: NSL-KDD: Confusion matrix visualization.

4.1.5 ROC Curve

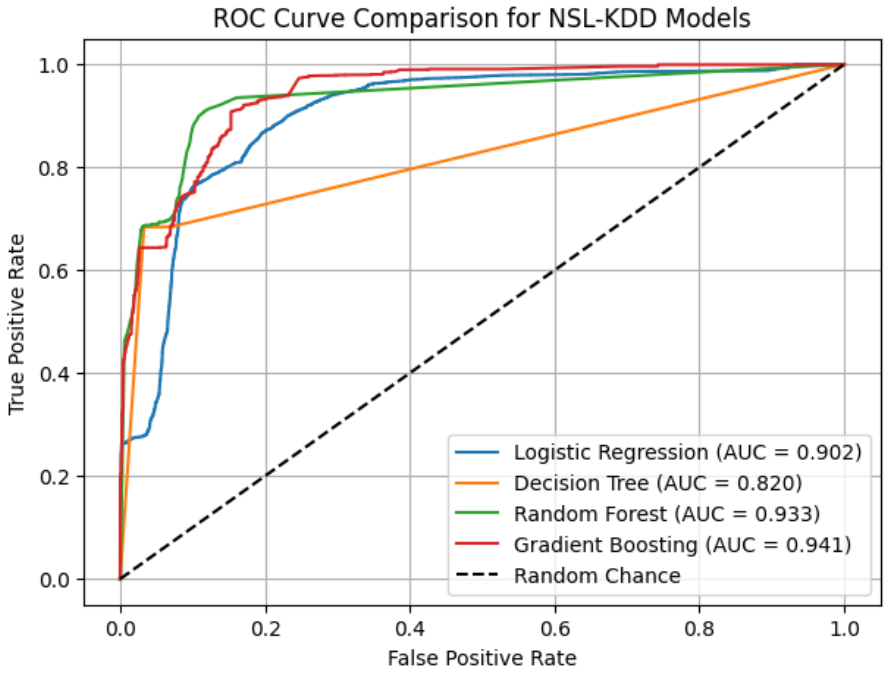

Figure 3: NSL-KDD: ROC curves for all four models.

4.1.6 Model Comparison Analysis

The NSL-KDD dataset shows that Random Forest achieves the best balance between accuracy and F1-score, while Gradient Boosting slightly underperforms compared to expectations. Logistic Regression remains interpretable but limited for complex non-linear attack detection.

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4.2 CICIDS2017 Results

4.2.1 Training vs Testing Accuracy

Table 7: CICIDS2017: Training and Testing Accuracies

Model	Training Accuracy (%)	Testing Accuracy (%)
Logistic Regression	96.92	97.04
Decision Tree	99.06	99.08
Random Forest	99.78	99.77
Gradient Boosting	99.83	99.83

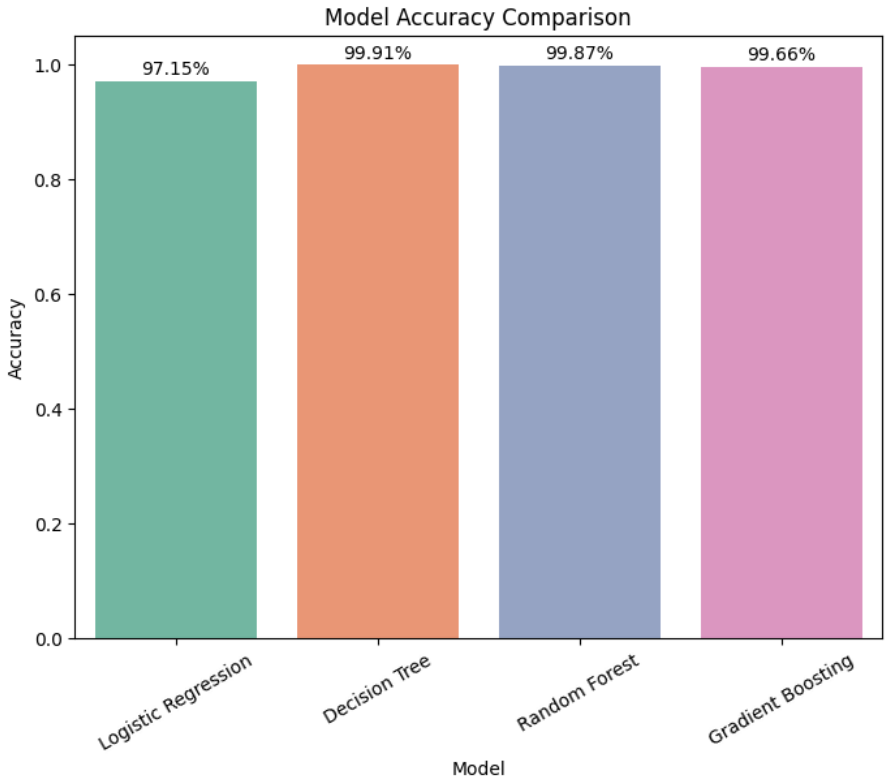

Figure 4: CICIDS2017: Model accuracy comparison.

4.2.2 Macro Average Performance

Table 8: CICIDS2017: Macro Average Precision, Recall, F1

Model	Precision	Recall	F1-score
Logistic Regression	0.96	0.97	0.97
Decision Tree	0.99	0.99	0.99
Random Forest	1.00	1.00	1.00
Gradient Boosting	1.00	1.00	1.00

4.2.3 Weighted Average Performance

Table 9: CICIDS2017: Weighted Average Precision, Recall, F1

Model	Precision	Recall	F1-score
Logistic Regression	0.97	0.97	0.97
Decision Tree	0.99	0.99	0.99
Random Forest	1.00	1.00	1.00
Gradient Boosting	1.00	1.00	1.00

4.2.4 Confusion Matrix

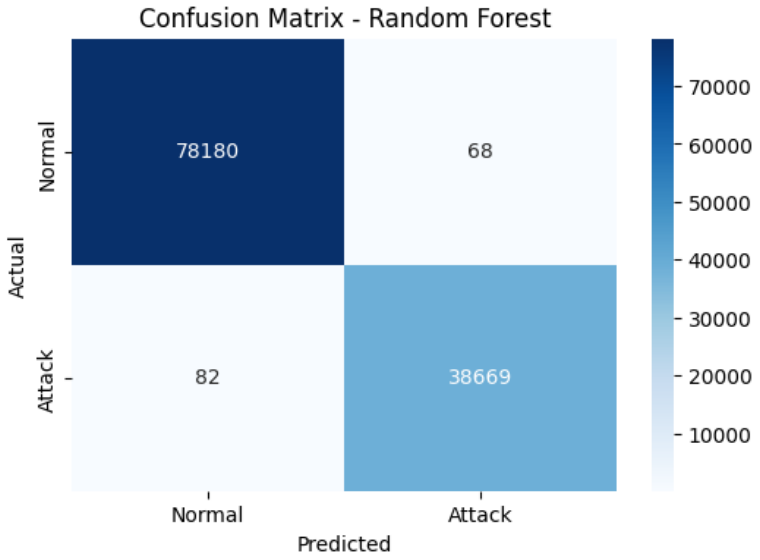

Figure 5: CICIDS2017: Confusion matrix visualization.

4.2.5 ROC Curve

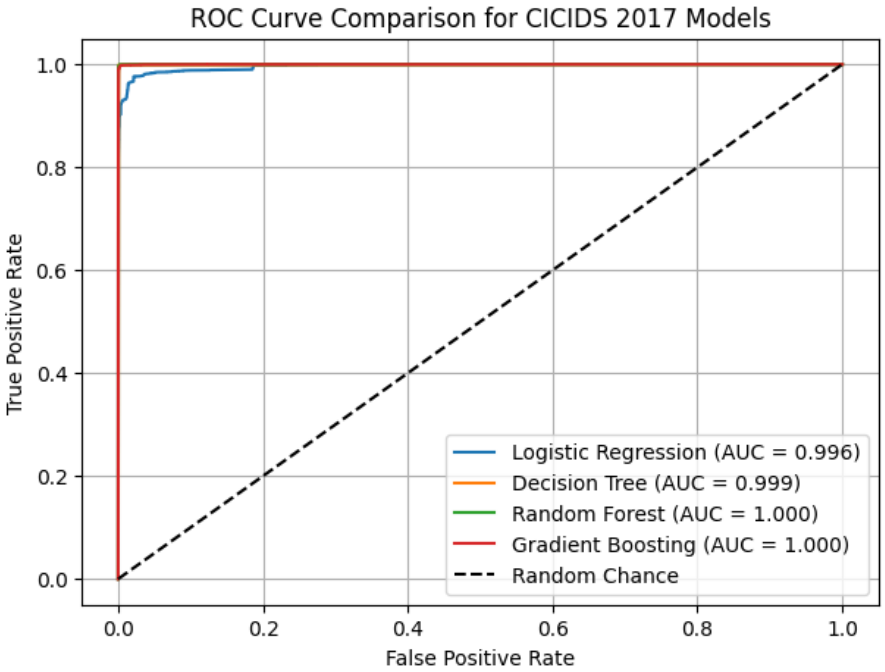

Figure 6: CICIDS2017: ROC curves for all four models.

4.2.6 Model Comparison Analysis

The CICIDS2017 dataset confirms that ensemble methods (Random Forest, Gradient Boosting) perform nearly perfectly, reflecting their robustness on modern attack traffic. Logistic Regression, while performing well, lags slightly due to its linear assumptions.

5 DISCUSSION

The results demonstrate that tree-based models like Random Forest and Gradient Boosting consistently outperform linear models such as Logistic Regression, both on NSL-KDD and CICIDS2017.

5.1 Key Insights

- NSL-KDD remains useful for academic benchmarking but lacks modern attack representation.
- CICIDS2017 better reflects real-world attack traffic, showing the potential of ML for deployment.
- Handling imbalance and rare classes remains critical.

6 CONCLUSION

SentinelNet demonstrates how AI-powered approaches can significantly enhance intrusion detection. The use of hybrid detection techniques, advanced preprocessing, and careful model evaluation allows for accurate and scalable intrusion detection systems. While results are promising, handling class imbalance and detecting low-frequency attack types remain open challenges.

7 FUTURE WORK

- Incorporating deep learning (CNNs, RNNs, Transformers) for sequential traffic patterns.
- Exploring online learning for real-time adaptation to evolving threats.
- Enhancing explainability of AI models to aid cybersecurity analysts.

- Extending evaluation to encrypted traffic analysis.
- Integrating SentinelNet with SIEM (Security Information and Event Management) systems.

8 ACKNOWLEDGEMENTS

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